Macromate

AI Project Proposal

Team Members

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1. Project Overview

Macromate is a small nutrition-logging prototype. A user uploads a photograph of one food item. An existing AI image model suggests a food name from a short list. The user checks the suggestion and enters the serving size in grams. The system then estimates calories, protein, carbohydrates, and fat and adds the food to a daily log.

The first version will support about six clearly named foods. Possible examples are apple, banana, orange, boiled egg, cooked rice, and sliced bread. The final list will depend on the images and nutrition records we can obtain. The system will not try to identify every possible meal or guess serving weight from a photograph.

2. Problem and Users

Recording a meal by hand takes time. A person must identify the food, search for nutrition information, and calculate the amount eaten. This repeated work can discourage regular tracking. Macromate aims to make the first step faster by suggesting the food from an image.

The intended users are adults tracking food for general fitness goals. The prototype is an academic demonstration. Its nutrition values are estimates and are not medical advice.

3. Why AI Is Needed

Food photographs vary in lighting, angle, background, and appearance. A simple list of rules cannot reliably recognize the food in every image. A pretrained image model can compare a photograph with the names of the supported foods and suggest the closest match. We plan to test a third-party model such as CLIP. We do not plan to train a large model from the beginning.

4. Proposed Prototype

The user will upload one image and see the model's suggested food. The user can correct the food name before saving it. The user will then enter the serving size in grams. For each food, a small reference table will store calories, protein, carbohydrates, and fat per 100 grams. For example, a 150-gram serving uses 1.5 times the values listed for 100 grams. The system will add the calculated values to the daily total.

Simple messages may compare the daily totals with the user's stated fitness goal. These messages will follow clear rules; they will not be presented as medical recommendations.

5. Data and Preparation

We plan to use about 20-30 labeled images for each of six food classes. We will record the image source and the correct food label for each file. A small nutrition table will be built from documented food records, such as USDA FoodData Central. Each nutrition entry will record its food name, amount per 100 grams, and source record.

Python scripts will check for images with missing labels, unreadable files, and duplicates. For nutrition records, they will check missing values, duplicates, and unusual values such as negative calories. The scripts will use consistent units. Images will be resized and normalized as required by the chosen model, and food labels will be encoded consistently. We will show simple charts of image counts by food class, missing data, and nutrition values.

6. How We Will Evaluate It

We will keep about 10 labeled images per food class separate for testing. We will report the percentage of correct predictions and show a few mistakes. We will also check the nutrition calculation by hand for different serving sizes. If the model suggests the wrong food, we will test whether the user can correct it before the entry is saved. The report will explain that mixed meals and different recipes remain outside the prototype's scope.